

```
from google.colab import files
```

```
uploaded = files.upload()
```

[Choose Files](#) ratings.csv

**ratings.csv**(text/csv) - 2483723 bytes, last modified: 8/17/2026 - 100% done

Saving ratings.csv to ratings.csv

```
import pandas as pd
```

```
filename = list(uploaded.keys())[0]
```

```
ratings = pd.read_csv(filename)
```

```
print("Dataset Shape:", ratings.shape)
```

```
display(ratings.head())
```

Dataset Shape: (100836, 4)

|   | userId | movieId | rating | timestamp |  |
|---|--------|---------|--------|-----------|--|
| 0 | 1      | 1       | 4.0    | 964982703 |  |
| 1 | 1      | 3       | 4.0    | 964981247 |  |
| 2 | 1      | 6       | 4.0    | 964982224 |  |
| 3 | 1      | 47      | 5.0    | 964983815 |  |
| 4 | 1      | 50      | 5.0    | 964982931 |  |

```
ratings = ratings[
    ["userId", "movieId", "rating"]
]
```

```
print("Records:", len(ratings))
```

```
display(ratings.head())
```

Records: 100836

|   | userId | movieId | rating |  |
|---|--------|---------|--------|--|
| 0 | 1      | 1       | 4.0    |  |
| 1 | 1      | 3       | 4.0    |  |
| 2 | 1      | 6       | 4.0    |  |
| 3 | 1      | 47      | 5.0    |  |
| 4 | 1      | 50      | 5.0    |  |

```
import numpy as np
```

```
from sklearn.neighbors import NearestNeighbors
```

```

movie_stats = (
    ratings
    .groupby("movieId")["rating"]
    .agg(["mean", "count"])
    .reset_index()
)

movie_stats.columns = [
    "movieId",
    "avg_rating",
    "n_ratings"
]

query = np.array([[4.0, 10.0]])

X = movie_stats[
    ["avg_rating", "n_ratings"]
].values

knn = NearestNeighbors(
    n_neighbors=5
)

knn.fit(X)

distances, indices = knn.kneighbors(query)

similar_movies = movie_stats.iloc[
    indices[0]
]

display(similar_movies)

cbl_prediction = (
    similar_movies["avg_rating"].mean()
)

print(
    "CBL Prediction:",
    round(cbl_prediction, 3)
)

```

|             | movieId | avg_rating | n_ratings |                                                                                     |
|-------------|---------|------------|-----------|-------------------------------------------------------------------------------------|
| <b>2193</b> | 2917    | 4.00       | 10        |  |
| <b>83</b>   | 94      | 3.95       | 10        |  |
| <b>887</b>  | 1185    | 3.95       | 10        |                                                                                     |
| <b>834</b>  | 1096    | 4.05       | 10        |                                                                                     |
| <b>939</b>  | 1241    | 4.05       | 10        |                                                                                     |

CBL Prediction: 4.0

Next steps: [Generate code with similar\\_movies](#) [New interactive sheet](#)

```

X = movie_stats[
    ["avg_rating", "n_ratings"]
].values

y = movie_stats[
    "avg_rating"
].values

distance = np.linalg.norm(
    X - query,
    axis=1
)

weights = np.exp(
    -(distance ** 2) / 2
)

A = np.column_stack([
    np.ones(len(X)),
    X
])

W = np.diag(weights)

theta = np.linalg.pinv(
    A.T @ W @ A
) @ A.T @ W @ y

new_point = np.array([
    1,
    4.0,
    10.0
])

lwr_prediction = (
    new_point @ theta
)

print(
    "LWR Prediction:",
    round(lwr_prediction, 3)
)

```

LWR Prediction: 4.0

```

import matplotlib.pyplot as plt

methods = [
    "Case-Based Learning",
    "LWR"
]

predictions = [
    cbl_prediction,
    lwr_prediction
]

```

```
plt.figure(figsize=(8,5))

plt.bar(
    methods,
    predictions
)

plt.ylabel("Predicted Rating")
plt.title(
    "Case-Based Learning vs LWR"
)

plt.ylim(0, 5)

plt.grid(
    axis="y",
    alpha=0.3
)

plt.show()
```



